

AUSTIN GIFFEN

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2024 U.S. Presidential Scholar, National Merit Scholar

EDUCATION

Yale University (GPA: 3.91)

Major: Electrical Engineering

New Haven, CT

Graduating Spring 2028

• Relevant Courses

- Probabilistic Machine Learning, Quantum Information Processing and Communication, Mechatronics Laboratory, Circuits and Systems Design, Digital Systems, Information Systems, Intensive Physics (classical mechanics/special relativity), Modern Physical Measurement, Linear Algebra, Computer Engineering, Intensive Physics (E&M/special relativity/quantum mechanics), Electromagnetic Waves and Devices, Electronics, Communications and Control

WORK EXPERIENCE

Avisitech, Inc | Electrical Engineering Intern

Summer 2026 + Fall 2026 Internships

- Developed analytical via-in-cavity EM models comparable to full-wave accuracy up to 50 GHz with orders-of-magnitude reduction in solve time.
- Validated formulations across canonical multi-layer PCB geometries using full-wave 3D planar MoM (Sonnet) and FDTD (openEMS) simulations.

Yale Undergraduate Research Assistant — Konezny Lab

January 2025 - January 2026

- Recipient of the Yale College First-Year Summer Research Fellowship (\$5,000 research grant)
- Designed and assembled gas flow control system using Arduino, mass flow controllers, and LabVIEW for precise hydrogen/oxygen doping experiments.
- Developed Python scripts to process experimental data, compute conductivity, and generate visual correlations.
- Reviewed and applied findings from academic literature to evaluate and optimize the conduction mechanisms of CuSCN

Yale Undergraduate Research Assistant — Jung Han Lab

January 2026 - May 2026

- Developed and optimized machine-learning models (Random Forest, cross-validated grid search) in Python to predict glucose levels from optical spectroscopy sensor signals.

Lead Sailing Instructor for Southern Idaho Sailing Association

May 2022 - August 2024

- Worked with two other instructors to run sailing camps for beginner through advanced sailors.

PROJECTS

Yale CubeSat (satellite) Project Lead

September 2024 - Present

- Manage a 10-person team, assigning project deliverables across subteams and leading NASA launch milestone meetings.
- Developed the complete cosmic ray detector schematic and PCB from negative voltage rail, multiple stage amplifiers, comparators, DACs, counter ICs, temperature sensors, boost converters, and multiplexers.
- Simulated circuits in KiCad using SPICE models for each component to validate gain, noise, timing response, and power behavior before fabrication.
- Tested and validated the detector subsystem through bench measurements, debugging, calibration, and data analysis from local cosmic-ray counts to confirm proper functionality.

Hardware-Aware Bayesian Neural Network Compiler Framework

Spring 2026

- Built a compiler framework to translate Bayesian Neural Networks (BNNs) into physical analog memristor crossbar circuits for low-power edge AI.
- Cut parameter search space by 50% by embedding real-world memristor physical constraints directly into the variational training objective.
- Minimized routing overhead and proved convergence, demonstrating that crossbar topology optimizes data movement while ensuring reliable probabilistic sampling under physical device noise.

MindVault: Consumer Learning App

Summer 2026 – Present

- Co-founded and built a consumer app that vaults lectures, articles, and videos, extracts them into discrete concepts, and models per-concept memory decay to serve personalized daily microquizzes.
- Managed market outreach and investment outreach, including creator partnership negotiations and a pilot program designed to validate user retention and engagement.

ADDITIONAL SKILLS

Engineering Skills: Circuit Design & Analysis, Oscilloscopes, Digital Multimeters, Waveform Generators, Verilog (RTL & Testbenches), Glove Box + Cryostat Operation, Spin Coating, Thin Film Deposition

Python Libraries: NumPy, Matplotlib, Scikit-learn, Pandas, tkinter

Workflow Tools: GitHub, Visual Studio Code, Spyder, Jupyter Notebook, Docker

Languages: English – Native, Spanish – Advanced: 2023 International Diploma of Spanish (D.I.E.)